AMIT KOTGI

amitamitkotgi@gmail.com | Belagavi, Karnataka
LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Electronics and Communication Engineering undergraduate at KLE Technological University, Belagavi (DCET lateral entry; expected graduation 2028), with hands-on experience in ESP32-based IoT prototypes, sensor interfaces, electronic circuits, telemetry, and web dashboards. Contributed to backend telemetry and dashboard development for GRIDGUARD AI, which secured Third Prize at KLE Tech HackFest 2026. Interested in developing reliable embedded and connected systems by combining firmware fundamentals, electronics knowledge, and practical problem-solving.

TECHNICAL SKILLS

Programming: C and Embedded C (fundamentals), Python, JavaScript (ES6), Arduino C++

Embedded & IoT: ESP32, Arduino, Wi-Fi telemetry, Firebase Realtime Database, ARM Cortex-M3 / LPC2148 (coursework)

Backend & Data: FastAPI, SQLite, REST APIs, WebSocket, Streamlit, LangGraph, Groq API

Electronics & Sensors: Op-amp circuits, comparators, signal conditioning, voltage/current sensing, ZMPT101B, ACS712, LM35, NTC thermistor, OLED, MPU6050

Tools & Web: Git/GitHub, VS Code, Arduino IDE, Keil, Proteus, LTspice, Multisim (basic), HTML5, CSS3, Chart.js, Tailwind CSS

Currently Learning: PID control, MATLAB/Simulink, FreeRTOS, Data Structures in C, VLSI fundamentals

PROJECTS

GRIDGUARD AI | Industrial Energy Monitoring & Anomaly Detection — Third Prize, KLE Tech HackFest 2026

Python, FastAPI, SQLite, WebSocket, ESP32, Chart.js | Team project; simulated prototype

- Implemented FastAPI telemetry ingestion and SQLite storage; contributed to the live monitoring dashboard using WebSocket and Chart.js.
- Demonstrated the monitoring pipeline with simulated telemetry from a low-voltage 12V DC motor; data was synthetic, not field data.

Smart Energy Meter IoT System | NMIT Hacks 2026 — IoT Track

ESP32, ZMPT101B, ACS712, OLED, Firebase

- Integrated voltage and current sensing with ESP32; displayed readings on an OLED and web dashboard and logged telemetry to Firebase.

CHITTI | Self-Balancing Two-Wheel Robot — In Development

ESP32, MPU6050, motor driver, embedded firmware

- Developing orientation sensing and motor control while exploring PID-based balance control; closed-loop balancing performance is not yet validated.

Portable Battery Condition Tester | Resonance Transistron 2026, KLE Tech

LM393 dual comparator, Arduino, LED indicators

- Designed and analysed a comparator circuit that indicates GOOD, WEAK, and DEAD battery states through LED outputs.

Solar Battery Management Dashboard | Simulated Prototype

HTML5, JavaScript, Tailwind CSS, Chart.js, Firebase

- Created a dashboard for voltage, current, state of charge, and temperature, with charts, an energy-flow view, alerts, and a relay-control interface; hardware integration remains planned.

BANKMATE AI | Banking Customer Query Assistant — TCS AI Hackathon

Python, LangGraph, Groq, FastAPI, Streamlit, SQLite | Four-member team

- Prototyped banking query flows during a 3.5-hour hackathon session; used synthetic data only and was not a live banking service.

EDUCATION

B.E. in Electronics and Communication Engineering — KLE Technological University, Dr. M. S. Sheshagiri Campus, Belagavi | DCET lateral entry | 5th Semester | Expected 2028

Diploma in Electronics and Communication Engineering — Nutan Vidyalaya Polytechnic, Kalaburagi | 2021–2024

ACHIEVEMENTS & COMPETITIONS

- Third Prize, KLE Tech HackFest 2026 (11 October 2026) — GRIDGUARD AI team project.
- Participant, TCS AI Hackathon — BANKMATE AI collaborative prototype.
- Participant, Resonance Transistron 2026 and Buildverse Challenge League (Pleiades Fest), KLE Tech.
- Registered, MathWorks Minidrone Competition India 2026 — preparing the Round 1 MATLAB/Simulink simulation.

RELEVANT COURSEWORK

Embedded Systems, ARM Architecture, Analog Electronics, Signals and Systems, Control Systems, Digital Signal Processing, CMOS VLSI, Machine Learning and Deep Learning, Operating Systems and Embedded Systems, Communication Systems, Probability and Random Processes, RTOS Lab.